Lab 5: Predicting telecom churn

70-467 - Machine Learning for Business Analytics

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2026-10-27

Background. A telecom company sells phone and internet plans on monthly bills. Every month some customers cancel and move to another provider. In the industry this is called churn. The retention department can contact some customers with an offer to stay, not everyone.

The dataset has 7,032 customers. 1,869 left (26.6 percent). Tenure runs from 1 to 72 months, and the monthly bill from $18.25 to $118.75. The other columns are contract, senior citizen, internet service, and payment method. The codebook on the next page gives the type and unit of every variable.

The problem. The retention department wants a list of customers to contact first. A model can give every customer a probability of churn. You still have to choose a threshold. Each threshold contacts some customers who would have stayed and misses some who leave. A model can also look accurate while missing most of the customers who leave, because most customers stay.

This is the question we asked in class: how do we turn what we know about a customer into a probability, how do we read that model, and how accurate are its predictions on customers it has not seen?

What we will do in the lab. We will use logistic regression from class.

First we will hold out 30 percent of the customers. We will estimate the model on the other 70 percent with four variables: tenure, monthly charges, contract, and senior citizen. We will read each coefficient as a change in the log-odds, a multiplier of the odds, and a change in the probability, for one customer and on average over all customers.

On the training customers we will read the confusion matrix, sensitivity, specificity, and precision, at a threshold of 0.50 and at 0.30, and we will build a table of thresholds. Each of you will choose one threshold from that training table and say why. On the held-out customers we will read the four counts for the threshold you chose, the same two thresholds for comparison, the receiver operating characteristic curve, the area under that curve, and lift. Internet service and payment method are for the questions after class.

Read the codebook before class. Bring a laptop with the packages from the starter page installed. After class you will have additional questions connected to the logistic regression slides.

Codebook. 7,032 observations, 7 variables. One observation = one customer. No missing values.

Variable	Type	Unit	Description
Churn	integer	0 / 1	1 if the customer left the company, 0 if the customer stayed. The outcome.
tenure	integer	months	Months the customer has been with the company.
MonthlyCharges	numeric	dollars	The customer's monthly bill.
Contract	text	none	Month-to-month, One year, or Two year.
SeniorCitizen	integer	0 / 1	1 if the customer is a senior citizen, 0 otherwise.
InternetService	text	none	DSL, Fiber optic, or No (no internet service). After class only.
PaymentMethod	text	none	Bank transfer, Credit card, Electronic check, or Mailed check. After class only.